

Lab 1A — AWS CLI Proof of Concept (No Bash Variables)

This document provides an AWS CLI-only proof of concept for Lab 1A. All commands are written as direct console commands with placeholders instead of bash variables. Replace placeholder values (e.g.,) with real AWS IDs before running.

Step 0 – Identify Account ID

```
aws sts get-caller-identity --query Account --output text
```

Step 1 – Create Security Groups (EC2 + RDS)

```
aws ec2 create-security-group --region us-east-1 --vpc-id <VPC_ID> --group-name sg-ec2-lab --description "EC2 Security Group"
aws ec2 authorize-security-group-ingress --region us-east-1 --group-id <EC2_SG_ID> --protocol tcp --port 22 --source <EC2_PUBLIC_IP>
aws ec2 authorize-security-group-ingress --region us-east-1 --group-id <EC2_SG_ID> --protocol tcp --port 80 --source <EC2_PUBLIC_IP>
aws ec2 create-security-group --region us-east-1 --vpc-id <VPC_ID> --group-name sg-rds-lab --description "RDS Security Group"
aws ec2 authorize-security-group-ingress --region us-east-1 --group-id <RDS_SG_ID> --ip-permissions '[{"IPProtocol": "tcp", "From": "0.0.0.0/0", "To": "0.0.0.0/0", "Ports": [3306]}']
```

Step 2 – RDS Subnet Group and Database

```
aws rds create-db-subnet-group --region us-east-1 --db-subnet-group-name lab-private-dbsubnets --db-subnet-ids <SUBNET_IDS>
aws rds create-db-instance --region us-east-1 --db-instance-identifier lab-mysql --engine mysql --db-instance-class db.t3.micro --availability-zone us-east-1a --vpc-subnet-ids <SUBNET_IDS>
aws rds wait db-instance-available --region us-east-1 --db-instance-identifier lab-mysql
aws rds describe-db-instances --region us-east-1 --db-instance-identifier lab-mysql --query "DBInstances[0].DBInstanceIdentifier"
```

Step 3 – Create Secrets Manager Secret

```
aws secretsmanager create-secret --region us-east-1 --name lab/rds/mysql --secret-string '{"username": "root", "password": "password"}'
```

Step 4 – IAM Role and Instance Profile

```
aws iam create-role --role-name lab-ec2-secrets-role --assume-role-policy-document file://trust.json
aws iam put-role-policy --role-name lab-ec2-secrets-role --policy-name lab-read-secret-inline --policy-document file://lab-read-secret-inline.json
aws iam create-instance-profile --instance-profile-name lab-ec2-instance-profile
aws iam add-role-to-instance-profile --instance-profile-name lab-ec2-instance-profile --role-name lab-ec2-secrets-role
```

Step 5 – Launch EC2 Instance with User Data

```
aws ec2 run-instances --region us-east-1 --image-id ami-07ff62358b87c7116 --instance-type t3.micro --subnets <SUBNET_IDS> --security-groups <SG_IDS> --user-data file://user-data.sh
```

Step 6 – Application Validation

```
http://<EC2_PUBLIC_IP>/init
http://<EC2_PUBLIC_IP>/add?note=first_note
http://<EC2_PUBLIC_IP>/list
```